Personal notes : Dynamical EIT cheat sheet

I. DYNAMICAL EIT PROTOCOL

Reminder : in all my notes I define the single collective excitation as such : for an ensemble of N atoms with a ground state $|g\rangle$ and an auxiliary state $|s\rangle$, it is written as:

$$|\Psi\rangle = \frac{1}{\sqrt{N}} \sum_{j=1}^{N} e^{i\phi_j} \left| g_1 \cdots s_j \cdots g_N \right\rangle, \tag{1}$$

where ϕ_j is a phase factor, left unspecified for now

Electromagnetically Induced Transparency (EIT) provides a coherent interface between propagating light fields and collective atomic excitations. It arises from destructive quantum interference between two excitation pathways in a multilevel atomic medium, resulting in a narrow spectral window of transparency within an otherwise absorbing resonance. Beyond its original interpretation as a suppression of absorption, EIT represents the foundation of many coherent light-matter mapping protocols, allowing the reversible conversion of optical fields into long-lived collective excitations. We show here how a single photon probe can be used to generate a collective excitation $|\Psi\rangle$.

We consider an ensemble of N identical atoms, each possessing two long-lived ground states $|g\rangle$ and $|s\rangle$, and a single optically excited state $|e\rangle$. The transition $|g\rangle \leftrightarrow |e\rangle$ is coupled to a weak quantized probe field, while $|s\rangle \leftrightarrow |e\rangle$ is driven by a strong classical control field. Under the rotating-wave approximation, the total interaction Hamiltonian can be written as

$$\hat{H}_{\text{int}} = -\hat{\mathbf{d}}^{(-)} \cdot \hat{\mathbf{E}}^{(+)}(\mathbf{r}) - \hat{\mathbf{d}}^{(-)} \cdot \mathbf{E}_{\text{c}}^{(+)}(\mathbf{r}, t) + \text{h.c.}, \tag{2}$$

where the dipole operator is $\hat{\mathbf{d}}^{(-)} = d_{eg} \sum_{j=1}^{N} \hat{\sigma}_j^{eg} + d_{es} \sum_{j=1}^{N} \hat{\sigma}_j^{es}$, and d_{eg} and d_{es} are the dipole moments for the respective transitions. The quantized probe field is expressed in the plane-wave basis as $\hat{\mathbf{E}}^{(+)}(\mathbf{r}) = \sum_{\mathbf{k}_p} \mathcal{E}_{\mathbf{k}_p} \hat{a}_{\mathbf{k}_p} e^{i\mathbf{k}_p \cdot \mathbf{r}}$, with $\mathcal{E}_{\mathbf{k}_p} = -i\sqrt{\hbar c k_p / (2\epsilon_0 \mathcal{V})}$, while the control field is treated classically as $\mathbf{E}_{\text{c}}^{(+)}(\mathbf{r}, t) = \mathcal{E}_{\text{c}}(t) e^{i\mathbf{k}_{\text{c}} \cdot \mathbf{r}}$. Substituting these expressions into Eq. (2) and collecting terms under the rotating-wave approximation yields

$$\hat{H}_{\text{EIT}} = \hbar \sum_{j=1}^{N} \left[g_{\mathbf{k}_p} \hat{a}_{\mathbf{k}_p} e^{i\mathbf{k}_p \cdot \mathbf{r}_j} \hat{\sigma}_j^{eg} + \Omega_{\text{c}}(t) e^{i\mathbf{k}_{\text{c}} \cdot \mathbf{r}_j} \hat{\sigma}_j^{es} + \text{h.c.} \right], \tag{3}$$

where $\hbar g_{\mathbf{k}_p} = d_{eg} \mathcal{E}_{\mathbf{k}_p}$ is the single-atom coupling constant for the probe mode, and $\hbar \Omega_c(t) = d_{es} \mathcal{E}_{\text{c}}(t)$ defines the (time-dependent) Rabi frequency of the control field.

We note that this Hamiltonian conserves the total number of excitations $\hat{N} = \hat{a}_{\mathbf{k}_p}^{\dagger} \hat{a}_{\mathbf{k}_p} + \sum_{j=1}^{N} \left(\hat{\sigma}_j^{ee} + \hat{\sigma}_j^{ss} \right)$, since $[\hat{H}_{\text{EIT}}, \hat{N}] = 0$. Therefore, the subspace with a single excitation $\hat{N} = 1$, is invariant under the dynamics. We now restrict this discussion to the single-excitation manifold, which is only valid if the incoming probe field contains a single photon at most. In this context, our new basis is made of $|1\rangle \equiv \left|1_{\mathbf{k}_p}\right\rangle \otimes |G\rangle$, $|\Psi\rangle \equiv \left|0_{\mathbf{k}_p}\right\rangle \otimes \hat{s}_{\mathbf{k}_{\text{sw}}}^{\dagger} |G\rangle$, and $|E\rangle \equiv \left|0_{\mathbf{k}_p}\right\rangle \otimes \hat{e}_{\mathbf{k}_p}^{\dagger} |G\rangle$, where the collective operators are defined as $\hat{e}_{\mathbf{k}_p}^{\dagger} = \sum_{j=1}^{N} e^{i\mathbf{k}_p \cdot \mathbf{r}_j} \hat{\sigma}_j^{eg} / \sqrt{N}$ and $\hat{s}_{\mathbf{k}_{\text{sw}}}^{\dagger} = \sum_{j=1}^{N} e^{i(\mathbf{k}_p - \mathbf{k}_c) \cdot \mathbf{r}_j} \hat{\sigma}_j^{sg} / \sqrt{N}$, with $\mathbf{k}_{\text{sw}} = \mathbf{k}_p - \mathbf{k}_c$ the spin-wave wavevector. We ignored other orthogonal Dicke states because, in the ideal case, they are not populated.

In the two-photon rotating frame (probe at $\mathbf{k}_p$ and control at $\mathbf{k}_c$), the couplings become spatially uniform and real. On two-photon resonance and within RWA, the full Hamiltonian projected onto $\{|1\rangle, |\Psi\rangle, |E\rangle\}$ reads, with $\Delta = \Delta_p = \Delta_c$ the detuning of each beam relative to the excited state:

$$\hat{H} = \hbar \begin{pmatrix} 0 & 0 & \sqrt{N}\, g_{\mathbf{k}_p} \\ 0 & 0 & \Omega_c(t) \\ \sqrt{N}\, g_{\mathbf{k}_p} & \Omega_c(t) & \Delta \end{pmatrix}. \tag{4}$$

The EIT condition is obtained for $\Delta = 0$. Diagonalizing eq. 4 in this context yields one dark state $|D\rangle$ orthogonal to $|E\rangle$, and two bright states that involve population in $|E\rangle$. The dark state is $|D\rangle = \cos\theta(t)\ |1\rangle - \sin\theta(t)\ |\Psi\rangle$, where $\tan\theta(t) = \frac{\sqrt{N}\, g_{\mathbf{k}_p}}{\Omega_c(t)}$. Indeed, $\langle E|D\rangle = 0$, so $|D\rangle$ is therefore dark to absorption and spontaneous emission.

To describe the propagation dynamics, we move from the state picture to the Heisenberg picture, where the corresponding field operators evolve in time. This leads to the so-called dark-state polariton operator describing the dynamical EIT process, that combines the probe envelope and the spin wave:

$$\hat{\Phi}(z,t) = \cos\theta(t)\,\hat{\mathcal{E}}(z,t) - \sin\theta(t)\,\sqrt{N}\,\hat{s}_{\mathbf{k}_{sw}}(z,t), \tag{5}$$

where the mixing angle

$$\tan\theta(t) = \frac{\sqrt{N}\,g_{\mathbf{k}_p}}{\Omega_c(t)}. \tag{6}$$

can be changed by controlling $\Omega_c(t)$.

Now, if the rate of change of the mixing angle is small compared to the energy splitting of the bright and dark state, there exists no coupling from the dark-state to the bright state (as detailed later in section II), so that the system remains in the dark manifold at all times. This condition is called the adiabatic condition, and can be expressed as $\left|\dot{\Omega}_c(t)\right| \ll g^2 N$. Besides, we assume that the optical polarization $\hat{\sigma}^{ge}$ is a fast variable that adiabatically follows the fields, yielding the effective propagation equation for the dark-state polariton (details in section III B):

$$\left[\frac{\partial}{\partial t} + v_g(t)\,\frac{\partial}{\partial z}\right]\hat{\Phi}(z,t) = 0, \tag{7}$$

with a group velocity

$$v_g(t) = c\,\cos^2\theta(t). \tag{8}$$

As $\Omega_c(t)$ is adiabatically reduced, $\cos\theta(t)$ decreases, slowing the polariton and converting its photonic part into a collective spin excitation. In the limit $\Omega_c \to 0$, $v_g \to 0$, and the excitation is stored entirely in the ensemble:

$$\hat{\Phi}(z,t) \xrightarrow[\Omega_c\to 0]{} -\sqrt{N}\,\hat{s}_{\mathbf{k}_{sw}}(z,t). \tag{9}$$

By employing dynamical EIT, we can therefore generate the collective excitation $|\Psi\rangle$, given that a single photon is available initially. Reapplying the control field retrieves the photon mode, since in the limit of $\theta \approx 0$ we have $\hat{\Phi}(z,t) \to \hat{\mathcal{E}}(z,t)$. This holds provided the phase-matching condition $\mathbf{k}_r = \mathbf{k}_p - \mathbf{k}_c$ is satisfied, dephasing is negligible and optical depth is large ($OD \propto N \gg 1$).

This reversible mapping between light and collective atomic excitation constitutes the core mechanism of EIT quantum memories. Note that adiabatic conditions must be maintained to avoid populating the excited state, leading to incoherent photon loss. Furthermore, other processes, such as off-resonant scattering with other excited states or even laser phase noise might reduce the overall transparency (see my thesis). Another condition which was not discussed here is that the photonic excitation spatial extent must fit entirely in the medium when the control beam is turned off (see my thesis).

II. ADIABATIC LIMIT IN DYNAMICAL EIT

We derive the adiabaticity condition ensuring that a system prepared in the dark state remains in the dark manifold during the time dependent variation of the control field. We work in the single excitation manifold with basis $\{|1\rangle, |\Psi\rangle, |E\rangle\}$ and Hamiltonian eq. 4 with $\Delta = 0$. With the mixing angle $\tan\theta(t) = \frac{\sqrt{N}\,g}{\Omega_c(t)}$, we define the dark and bright states

$$|D(t)\rangle = \cos\theta(t)\,|1\rangle - \sin\theta(t)\,|\Psi\rangle, \tag{10}$$

$$|B(t)\rangle = \sin\theta(t)\,|1\rangle + \cos\theta(t)\,|\Psi\rangle, \tag{11}$$

$$|E\rangle = |E\rangle. \tag{12}$$

For fixed θ, one verifies

$$H\,|D\rangle = 0, \qquad H\,|B\rangle = \hbar\Omega_{\text{eff}}\,|E\rangle, \qquad H\,|E\rangle = \hbar\Omega_{\text{eff}}\,|B\rangle, \tag{13}$$

where

$$\Omega_{\text{eff}}(t) = \sqrt{Ng^2 + |\Omega_c(t)|^2} \tag{14}$$

is the generalized Rabi frequency. The instantaneous eigenvalues are $E_D = 0$ and $E_\pm = \pm\hbar\Omega_{\text{eff}}$. We expand the state vector as

$$|\psi(t)\rangle = c_D(t)\,|D(t)\rangle + c_B(t)\,|B(t)\rangle + c_E(t)\,|E\rangle, \tag{15}$$

and impose the Schrödinger equation:

$$\frac{d}{dt}|\psi\rangle = \dot{c}_D|D\rangle + \dot{c}_B|B\rangle + \dot{c}_E|E\rangle + c_D\left|\dot{D}\right\rangle + c_B\left|\dot{B}\right\rangle. \tag{16}$$

From the definitions of $|D\rangle$ and $|B\rangle$ one finds

$$\left|\dot{D}\right\rangle = \dot{\theta}(t)\left(-\sin\theta\ |1\rangle - \cos\theta\ |\Psi\rangle\right) = -\dot{\theta}(t)\ |B(t)\rangle, \tag{17}$$

$$\left|\dot{B}\right\rangle = \dot{\theta}(t)\left(\cos\theta\ |1\rangle - \sin\theta\ |\Psi\rangle\right) = \dot{\theta}(t)\ |D(t)\rangle, \tag{18}$$

and $\left|\dot{E}\right\rangle = 0$. Projecting the Schrödinger equation onto $\langle D|$, $\langle B|$ and $\langle E|$ yields:

$$i\hbar\,\dot{c}_D + i\hbar\,c_B\ \left\langle D\middle|\dot{B}\right\rangle = 0, \tag{19}$$

$$i\hbar\,\dot{c}_B + i\hbar\,c_D\ \left\langle B\middle|\dot{D}\right\rangle = \hbar\Omega_{\text{eff}}\,c_E, \tag{20}$$

$$i\hbar\,\dot{c}_E = \hbar\Omega_{\text{eff}}\,c_B. \tag{21}$$

Using $\left\langle D\middle|\dot{B}\right\rangle = \dot{\theta}$ and $\left\langle B\middle|\dot{D}\right\rangle = -\dot{\theta}$, we obtain the coupled equations

$$\dot{c}_D = -\dot{\theta}(t)\,c_B, \tag{22}$$

$$i\dot{c}_B - i\dot{\theta}(t)\,c_D = \Omega_{\text{eff}}(t)\,c_E, \tag{23}$$

$$i\dot{c}_E = \Omega_{\text{eff}}(t)\,c_B. \tag{24}$$

We now define a small dimensionless parameter

$$\varepsilon \equiv \frac{|\dot{\theta}(t)|}{\Omega_{\text{eff}}(t)} \ll 1. \tag{25}$$

and seek a perturbative solution in ε of the form

$$c_D(t) = c_D^{(0)}(t) + \varepsilon c_D^{(1)}(t) + \varepsilon^2 c_D^{(2)}(t) + \dots, \tag{26}$$

$$c_B(t) = \varepsilon c_B^{(1)}(t) + \varepsilon^2 c_B^{(2)}(t) + \dots, \tag{27}$$

$$c_E(t) = \varepsilon c_E^{(1)}(t) + \varepsilon^2 c_E^{(2)}(t) + \dots. \tag{28}$$

Just before the control pulse is turned on, the only populated state is the dark-state since the other states contain excited state population and decay rapidly. The initial condition is thus $c_D(t_0) = 1$, $c_B(t_0) = c_E(t_0) = 0$. Then, developing at first order in ε gives

$$\dot{c}_D^{(1)} = -\dot{\theta}(t)\,c_B^{(0)}(t) = 0, \tag{29}$$

$$i\,\dot{c}_B^{(1)} - i\,\dot{\theta}(t)\,c_D^{(0)}(t) = \Omega_{\text{eff}}(t)\,c_E^{(1)}(t), \tag{30}$$

$$i\,\dot{c}_E^{(1)} = \Omega_{\text{eff}}(t)\,c_B^{(1)}(t). \tag{31}$$

Equation (29) and $c_D^{(1)}(t_0) = 0$ imply $c_D^{(1)}(t) \equiv 0$, so the leading correction to c_D is of order ε^2. Eliminating $c_B^{(1)}$ by differentiating Eq. (31) and substituting Eq. (30) yields

$$\ddot{c}_E^{(1)}(t) + \Omega_{\text{eff}}^2(t)\,c_E^{(1)}(t) = i\,\Omega_{\text{eff}}(t)\,\dot{\theta}(t). \tag{32}$$

This is the equation of a driven harmonic oscillator. Using our initial conditions $c_E^{(1)}(t_0) = \dot{c}_E^{(1)}(t_0) = 0$, the particular solution has amplitude

$$c_E^{(1)}(t) \sim \frac{\dot{\theta}(t)}{\Omega_{\text{eff}}(t)}. \tag{33}$$

Substituting into Eq. (31) gives the same scaling for $c_B^{(1)}$. Collecting terms up to first order in ε yields

$$c_D(t) = 1 + \mathcal{O}(\varepsilon^2), \qquad c_B(t), c_E(t) = \mathcal{O}(\varepsilon) = \mathcal{O}\left(\frac{\dot{\theta}(t)}{\Omega_{\text{eff}}(t)}\right). \tag{34}$$

Thus suppressing population leakage out of the dark state requires

$$|\dot{\theta}(t)| \ll \sqrt{Ng^2 + |\Omega_c(t)|^2}. \tag{35}$$

Introducing a characteristic time scale T for the acceleration/deceleration period, Fleischhauer and Lukin derive an adiabatic condition related to experimental parameters as

$$T \gg \frac{Lv_g}{c^2}\frac{1}{\mathrm{OD}}. \tag{36}$$

For realistic experimental parameters, this is roughly $T \gg \mathcal{O}(10^{-16}\mathrm{s})$ which is largely fulfilled in practice since experiments using dynamical EIT are usually limited by acousto-optics modulator switching times of the order of the nanosecond.

III. DARK-STATE POLARITON PROPAGATION

A. Maxwell Bloch equations reminder

We consider an ensemble of N three-level atoms in a Λ configuration, interacting with a weak quantum signal field $\hat{\mathcal{E}}(z,t)$ and a strong classical control field with Rabi frequency $\Omega_c(t)$. After transforming to a slowly varying envelope picture and neglecting fast optical oscillations, the dynamics of the coupled atom-light system in the Heisenberg picture is governed by the Maxwell-Bloch equations.

The probe field satisfies the slowly-varying Maxwell equation

$$(\partial_t + c\,\partial_z)\,\hat{\mathcal{E}}(z,t) = ig\sqrt{N}\,\hat{\sigma}^{ge}(z,t), \tag{37}$$

where g is the single-photon Rabi frequency and $\hat{\sigma}^{\mu\nu}(z,t) \equiv \frac{1}{N_z}\sum_{j\in z}|\mu\rangle_j\langle\nu|$ denotes the collective atomic operators.

The atomic coherences evolve according to the Bloch equations

$$\partial_t\hat{\sigma}^{ge}(z,t) = -(\gamma + i\Delta)\,\hat{\sigma}^{ge}(z,t) + ig\,\hat{\mathcal{E}}(z,t) + i\Omega_c(t)\,\hat{\sigma}^{gs}(z,t), \tag{38}$$

$$\partial_t\hat{\sigma}^{gs}(z,t) = i\Omega_c^*(t)\,\hat{\sigma}^{ge}(z,t), \tag{39}$$

where Δ is the one-photon detuning and γ is the optical decoherence rate. These three equations constitute a general Maxwell-Bloch model for Λ type light-matter interfaces, including both EIT and Raman-type memories.

B. Dark-state polariton propagation

In this note we provide only the minimal steps required to obtain the propagation equation of the dark-state polariton. We start from the Maxwell-Bloch equations introduced in the previous section and use the same definitions of the slowly-varying probe field $\hat{\mathcal{E}}(z,t)$ and spin coherence $\hat{\sigma}^{gs}(z,t)$.

The excited-state coherence $\hat{\sigma}^{ge}$ obeys equation eq. 38. Because the excited state relaxes on the short timescale γ^{-1}, which is much faster than the characteristic variation of the probe field and the control-field envelope, the optical polarization adiabatically follows the fields. Therefore, we may neglect the time varying term and set $\partial_t\hat{\sigma}^{ge} = 0$. Using the Bloch equation for $\hat{\sigma}^{ge}$ then yields

$$\hat{\sigma}^{ge}(z,t) \simeq \frac{1}{\gamma + i\Delta}\left(g\,\hat{\mathcal{E}}(z,t) + \Omega_c(t)\,\hat{\sigma}^{gs}(z,t)\right). \tag{40}$$

The expression in parentheses is proportional to the bright polariton

$$\hat{\Xi} = \sin\theta\,\hat{\mathcal{E}} + \cos\theta\sqrt{N}\,\hat{\sigma}^{gs}, \tag{41}$$

with $\tan\theta = g\sqrt{N}/\Omega_c$. Since $g\sqrt{N}\cos\theta = \Omega_c\sin\theta$, one finds

$$\hat{\sigma}^{ge}(z,t) \simeq \frac{1}{\gamma + i\Delta}\frac{g}{\sin\theta}\,\hat{\Xi}. \tag{42}$$

In the ideal adiabatic condition, the bright polariton is never populated $\hat{\Xi} \simeq 0$, which means that

$$\hat{\sigma}^{ge}(z,t) \simeq 0 \tag{43}$$

and the excited state coherence $\hat{\sigma}^{ge}$ is removed from the equations of motion for $\hat{\Phi}$.

Taking the time derivative of the dark state polariton operator and substituting the Maxwell equation for $\hat{\mathcal{E}}$ and the spin-coherence equation for $\hat{\sigma}^{gs}$ gives

$$\begin{aligned}\partial_t\hat{\Phi} = \dot{\theta}\left[-\sin\theta\,\hat{\mathcal{E}} - \cos\theta\sqrt{N}\,\hat{\sigma}^{gs}\right] \\ + \cos\theta\left(-c\,\partial_z\hat{\mathcal{E}} + ig\sqrt{N}\,\hat{\sigma}^{ge}\right) \\ - \sin\theta\sqrt{N}\left(i\Omega_c^*\hat{\sigma}^{ge}\right).\end{aligned} \tag{44}$$

All terms involving $\hat{\sigma}^{ge}$ cancel. Neglecting non-adiabatic contributions proportional to $\dot{\theta}$, we obtain

$$\partial_t\hat{\Phi}(z,t) = -c\,\cos^2\theta(t)\,\partial_z\hat{\Phi}(z,t). \tag{45}$$

Thus the DSP obeys the simple first-order propagation equation

$$\left[\partial_t + v_g(t)\,\partial_z\right]\hat{\Phi}(z,t) = 0, \tag{46}$$

with the group velocity

$$v_g(t) = c\,\cos^2\theta(t) = c\,\frac{|\Omega_c(t)|^2}{|\Omega_c(t)|^2 + g^2N}. \tag{47}$$

This equation shows that, under adiabatic conditions, the dark state polariton propagates without distortion and with a controllable velocity set by the control field amplitude.

IV. EFFECT OF LASER PHASE NOISE AND LINEWIDTH

So far, we have assumed perfectly monochromatic probe and control fields. In practice, both optical fields exhibit phase fluctuations that can be written as $\phi_p(t)$ and $\phi_c(t)$, that are stochastic phases describing the finite laser linewidths. We have seen that, every atom, when the spin-wave is generated, grab a phase $e^{i(\mathbf{k}_p-\mathbf{k}_c).\mathbf{r}_j}$. But we assumed the global phase of both lasers to be stable over time. Now, if the global phase of the optical fields fluctuates in time, the relative phase between probe and control becomes time dependent. The process that generates the spin wave then transfers this relative phase to the collective excitation. The phase imprinted in Eq. (1) is thus $\phi_j = (\mathbf{k}_p - \mathbf{k}_c)\cdot\mathbf{r}_j + \Delta\phi(t_w)$, with $\Delta\phi(t_w) = \phi_p(t_w) - \phi_c(t_w)$ and t_w is defined as the time at which the photon was absorbed.

The probe pulse used to write the excitation has a finite temporal duration τ_{pulse}. Different temporal slices of the pulse are mapped into the atomic ensemble at different times. Each slice therefore carries the instantaneous relative phase of the probe and control fields at the moment of the mapping. If the relative phase evolves slowly compared to τ_{pulse}, all temporal components of the pulse imprint essentially the same global phase, and the collective excitation remains well defined. However, if the relative phase fluctuates significantly during the duration of the pulse, different parts of the pulse are written with different global phases. The atoms then acquire phases that vary randomly in time during the writing process. In this regime, the phase factors ϕ_j appearing in Eq. (1) are random across the ensemble, so the phase-matching condition required for efficient retrieval is lost.

Experimentally, the stability of the relative phase between the probe and control fields is characterized by the linewidth of their optical beat note. If the two beams are overlapped on a fast photodiode, the resulting radio-frequency signal at frequency $\omega_p - \omega_c$ has a finite spectral width. This width is called the beat-note linewidth and directly measures how fast the relative optical phase fluctuates. The relevant comparison is therefore between the beat-note linewidth $\Delta\nu_{\text{beat}}$ and the inverse duration of the probe pulse. If the relative phase evolves on a time scale much longer than the pulse duration,

$$\Delta\nu_{\text{beat}}\,\tau_{\text{pulse}} \ll 1, \tag{48}$$

the relative phase remains essentially constant while the spin wave is written and the phase matching is preserved. Conversely, if the beat-note linewidth is large enough that the relative phase changes significantly during the pulse, the phases imprinted in the ensemble become effectively random and the collective interference is destroyed.

For this reason, experiments usually derive the probe and control fields from the same laser source, or actively phase-lock the two lasers. A good thing to keep in mind is that, without phase lock loop, the linewidth of the beatnote is the sum of both laser linewidth: $\Delta\nu_{\text{beat}} = \Delta\nu_{\text{p}} + \Delta\nu_{\text{c}}$.